

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Theoretical probability

1. Assertion (A): When one number is picked at random from the numbers 1 to 20, the probability that it is a number from 1 to 20 is 1. Reason (R): An event E with $P(E) = 1$ is a sure event. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2025, 1 mark**

SOLUTION

Every possible outcome is a number from 1 to 20, so the event is certain to happen and its probability is $\frac{20}{20} = 1$. A is true.

R is true, and it is exactly the reason for A.

Answer: (a)

2. The even numbers are removed from the list 1, 4, 7, 9, 16, 21, 25, and one of the remaining numbers is picked at random. Find the probability that it is prime. (a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{1}{7}$ (d) $\frac{2}{7}$ **CBSE 2024, 1 mark**

SOLUTION

Removing the even numbers 4 and 16 leaves 1, 7, 9, 21, 25: five numbers.

The only prime among them is 7 (1 is not prime; $9 = 3^2$, $21 = 3 \times 7$, $25 = 5^2$).

Probability = $\frac{1}{5}$

Answer: (b)

2 Two dice

3. Two dice are thrown together. Find the probability that the two numbers differ by 2. **CBSE 2025, 3 marks**

SOLUTION

There are $6 \times 6 = 36$ equally likely outcomes.

Pairs that differ by 2: (1, 3), (3, 1), (2, 4), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4), which is 8 outcomes.

Probability = $\frac{8}{36}$

Answer: $\frac{2}{9}$

4. Two dice are thrown together. Find the probability that the two numbers add up to 2, 3 or 5. (a) $\frac{7}{36}$ (b) $\frac{11}{36}$ (c) $\frac{5}{36}$ (d) $\frac{4}{9}$ **CBSE 2024, 1 mark**

SOLUTION

Sum 2: (1, 1). Sum 3: (1, 2), (2, 1). Sum 5: (1, 4), (2, 3), (3, 2), (4, 1).

Favourable outcomes = $1 + 2 + 4 = 7$ out of 36.

Answer: (a)

3 Playing cards

5. One card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is a red face card. **CBSE 2025, 1 mark**

SOLUTION

The face cards are the jack, queen and king. The two red suits (hearts and diamonds) have $2 \times 3 = 6$ of them.

$$\text{Probability} = \frac{6}{52}$$

Answer: $\frac{3}{26}$

6. A black card is missing from a pack of 52 playing cards. One card is drawn at random from the rest. Find the probability that it is the queen of hearts. **CBSE 2024, 2 marks**

SOLUTION

One black card is lost, so 51 cards remain.

The queen of hearts is a red card, so it is still in the pack.

Answer: $\frac{1}{51}$

4 Complementary events

7. A player wins a game with probability 0.79. What is the probability that the player loses it? (a) 1.79 (b) 0.31 (c) 0.21% (d) 0.21 **CBSE 2024, 1 mark**

SOLUTION

$$P(\text{losing}) = 1 - P(\text{winning}) = 1 - 0.79 = 0.21$$

Answer: (d)

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